Course Name:	SDT	**Semester:**	7
Date of Performance:	**28 / 07 / 2026**	**DIV/ Batch No:**	A
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TITLE: Problem Framing & POV Statements

Objectives:

This experiment will help students to understand the principles and importance of problem framing in design thinking by creating Point-of-View (POV) statements that capture user needs and insights. This will also introduce different POV techniques to software development contexts for defining actionable problem statements

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Expected OUTCOME of Experiment: CO 1 and CO 2

Books/ Journals/ Websites referred:

Stanford d.school – An Introduction to Design Thinking: PROCESS GUIDE Stanford University, Hasso Plattner Institute of Design.
Interaction Design Foundation – Design Thinking - Interaction Design Foundation. Interaction Design Foundation – Design Thinking
IDEO Design Thinking - IDEO – Design Thinking and Human-Centered Design resources. IDEO – Design Thinking
Brown, T. – Change by Design: How Design Thinking Creates New Alternatives for Business and Society - Harper Business, 2009.

< The students should mention clearly the books, websites or journal papers, and articles they have refereed to while completing the task>

Background Theory:

Understanding Problem Framing Problem framing is the process of defining a challenge in a way that guides creative exploration. Instead of jumping directly to solutions, framing ensures that teams pause to ask: *Are we solving the right problem?* This step is crucial in design thinking because poorly framed problems often lead to solutions that miss the mark.

Human-Centered Perspective In design thinking, framing is always human-centered. It draws from empathy research — interviews, observations, and user data — to highlight real needs rather than assumptions. By reframing problems around user experiences, teams uncover opportunities that traditional requirement gathering might overlook.

Characteristics of Effective Frames A good problem frame is broad enough to allow creativity but specific enough to provide direction. It avoids technical jargon and instead focuses on user goals, frustrations, and aspirations. For example, instead of framing a problem as *"Improve the login system"*, a human-centered frame would be *"Help students access learning resources quickly without frustration."*

Point-of-View (POV) Statements POV statements are concise articulations of user needs and insights. They typically follow the format: *[User] needs [Need] because [Insight]*. This

structure ensures clarity and helps teams stay focused on solving problems that matter. POV statements act as a bridge between empathy research and ideation.

Role in Software Development In software design, POV statements translate user research into actionable design goals. They guide developers and designers in prioritizing features, shaping user interfaces, and ensuring usability. By grounding requirements in POV statements, teams reduce the risk of building technically sound but user-unfriendly systems.

Iterative Nature and Impact Problem framing and POV statements are not static. As new insights emerge, they evolve to reflect changing user contexts. This iterative nature keeps the design process adaptive and responsive. Ultimately, effective framing and POV statements foster innovation, ensuring that solutions are desirable, feasible, and viable.

Creation of POV using User + Need + Insight

1. **User** The "User" component of a POV statement identifies the specific person or group for whom the design solution is intended. It is not a generic audience but a clearly defined persona based on empathy research. For example, instead of saying "students," a well-defined user might be "a first-year international student unfamiliar with the campus." This level of specificity ensures that the design team focuses on real individuals with unique contexts, rather than abstract categories.

2. **Need** The "Need" represents what the user is trying to accomplish or the challenge they face. Needs are expressed as verbs rather than nouns, emphasizing action and purpose. For instance, a student may need "to find accessible routes across campus quickly" rather than simply "a map." Framing needs in this way highlights the underlying goals and motivations, making it easier to design solutions that address the root problem rather than superficial symptoms.

3. **Insight** The "Insight" explains why the need exists and provides a deeper understanding of the user's behavior, emotions, or constraints. Insights are derived from empathy work — interviews, observations, and user research — and often reveal hidden pain points or opportunities. For example, the insight might be "because traditional campus maps are cluttered and do not highlight wheelchair-friendly paths." This contextual explanation

transforms the POV statement into a powerful design tool, guiding teams toward solutions that are both innovative and human-centered.

A software is to be designed for a campus visitor. There are various types of users and each one will have different expectations (point of view) from the software.

Sample POV Statements for College Campus Tour System

- **Prospective Student POV** *A prospective student needs an interactive campus tour system because navigating a large campus for the first time is confusing and overwhelming.*
- **Parent POV** *A visiting parent needs clear directions and contextual information because they want to ensure their child's safety and understand the facilities available.*
- **International Student POV** *An international student needs multilingual support in the campus tour system because language barriers make it difficult to fully understand the campus environment.*
- **Disabled Student POV** *A student with mobility challenges needs accessibility-focused navigation because traditional tours often overlook wheelchair-friendly routes and facilities.*
- **Alumni Visitor POV** *An alumnus revisiting the campus needs a personalized tour highlighting new developments because they want to reconnect with familiar spaces while exploring recent changes*

For each of these users and their respective POVs Statements could be

Type of User	**User**	**Need**	**Insight**
Prospective Student	A first-year prospective student visiting campus	Needs a guided way to explore facilities and academic departments	Because navigating a large campus alone is overwhelming and they want to feel confident about their choice of college
P**OV Statement:** A prospective student needs a guided way to explore facilities because navigating a large campus alone is overwhelming and impacts their decision-making.			
Parent Visitor	A parent accompanying their child on a campus tour.	Needs clear information about safety, amenities, and student support services.	Because parents are concerned about their child's well-being and want reassurance about the environment.
POV Statement: A parent needs clear information about safety and support services because they want reassurance about their child's well-being on campus.			

International Student	An international student unfamiliar with local language and culture.	Needs multilingual support and cultural context during the tour.	Because language barriers and cultural differences make it difficult to fully understand the campus environment.
POV Statement: An international student needs multilingual support during the tour because language barriers prevent them from fully engaging with the campus experience.			
Disabled Student	A student with mobility challenges.	Needs accessibility-focused navigation highlighting wheelchair-friendly routes.	Because traditional tours often overlook accessibility needs, making it difficult to participate fully
POV Statement: A student with mobility challenges needs accessibility-focused navigation because traditional tours overlook wheelchair-friendly routes and facilities.			
Alumni Visitor	An alumnus revisiting the campus.	Needs a personalized tour highlighting new developments and familiar landmarks.	Because they want to reconnect with their past experiences while exploring recent changes
POV Statement: An alumnus needs a personalized tour highlighting new developments because they want to reconnect with familiar spaces while discovering what has changed			

Now after these POVs are thought of the next step would be creating Empathy maps for each type of users.

The sample questions/ empathy maps could be:

Type Of User	Says	Thinks	Does	Feels
Prospective Student	What to tell others about the first impression of the campus?	What concerns run through mind when exploring the campus alone?	How to navigate currently or explore new places on campus?	How do you feel when you get lost or cannot find a facility?
Parent Visitor	What do you ask tour guides or staff about safety and amenities?	What worries do you have about your child's life on campus?	How do you check for facilities like hostels, medical centers, or security?	How do you feel when information about student support services is unclear?
International Student	What do you say when you struggle with language or cultural differences?	What thoughts come to mind when instructions are only in one language?	How do you try to overcome language barriers during the tour?	How do you feel when you cannot fully understand the campus environment?
Disabled Student	What do you mention about accessibility challenges during tours?	What thoughts arise when you see routes or facilities that are not accessible?	How do you navigate the campus when routes are not wheelchair-friendly?	How do you feel when accessibility needs are overlooked in campus tours?
Alumni Visitor	What do you say when you revisit familiar landmarks or see new developments?	What thoughts come to mind about how the campus has changed since you studied here?	How do you explore both old and new areas of the campus during your visit?	How do you feel when reconnecting with past experiences while discovering changes?

More detailed version could be

Type of User	Says	Thinks	Does	Feels
Prospective Student	"The campus is so big, I don't know where to start." / "I want to see labs and library."	"Will I be able to find my classes easily?" / "I hope facilities match what I saw online."	Walks with a map or phone; asks peers for directions; takes photos of landmarks.	Excited about opportunities but nervous and overwhelmed by campus size.
Parent Visitor	"Is the campus safe for my child?" /	"Will my child be comfortable	Checks hostel, medical center,	Concerned about safety but

	"Where are the medical and hostel facilities?"	and secure here?" / "How reliable is student support?"	and security posts; asks staff questions.	reassured when facilities are explained clearly.
International Student	"I can't understand everything; is there a guide in my language?"	"It's hard to follow instructions when they're only in English."	Uses translation apps; relies on peers or staff for explanations.	Anxious about missing details; relieved when multilingual support is available.
Disabled Student	"Are there wheelchair-friendly routes?" / "I struggle with stairs and narrow paths."	"Will I be able to attend classes without difficulty?"	Avoids inaccessible areas; seeks alternative routes; asks for assistance.	Frustrated when accessibility is overlooked; confident when inclusive facilities are highlighted.
Alumni Visitor	"This place has changed so much since I studied here."	"I wonder what new facilities have been added."	Visits old landmarks; explores new buildings; takes photos to compare past and present.	Nostalgic about familiar spaces; excited to see new developments

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Laboratory Work:

You are required to continue with the topic selected in Experiment No.1 and complete the following tasks.

Task 1: Problem Framing Exercise

- Collect user research data (interviews, surveys, observations). You would be having the empathy map created for each type of user.

- Identify recurring expectations and pain points (difficulties faced by them).
- Frame the problem in natural language terms; identify the needs .

1. User Research

User research was considered for two major user groups: busy working professionals and senior citizens. The research focuses on their expectations, difficulties, behaviors, and experiences while booking hospital appointments.

User Group 1: Busy Working Professionals

Busy working professionals generally have limited free time because of their work schedules. They prefer using online appointment systems because they want to avoid visiting hospitals or waiting in long queues.

User Expectations

- Quick appointment booking
- Real-time doctor availability
- Instant appointment confirmation
- Easy rescheduling
- Doctor ratings and reviews
- Secure online payment
- Appointment reminders
- Short and simple booking process

Pain Points

- Limited appointment slots
- Long booking procedures
- Appointment cancellations
- Slow application performance
- Lack of real-time availability
- Difficulty rescheduling appointments
- Unclear confirmation status
- Long waiting times

User Need

A busy working professional needs a fast, reliable, and efficient appointment booking system that allows them to book and manage appointments without spending too much time.

Problem Frame

Busy working professionals need a hospital appointment system that allows them to quickly find available doctors, book appointments, receive instant confirmation, and reschedule appointments because their work commitments leave them with limited time and unexpected schedule changes can make traditional appointment booking difficult.

This is consistent with the problem identified in Experiment 1, where the professional user requires quick booking, real-time availability, and instant confirmation.

User Group 2: Senior Citizens

Senior citizens may face difficulties while using complicated digital applications. They prefer simple interfaces, readable text, clear navigation, and assistance when performing important tasks such as booking appointments and making payments.

User Expectations

- Large and readable text
- Simple navigation
- Clear buttons and instructions
- Easy appointment booking
- Voice assistance
- Easy access to prescriptions and medical records
- Simple payment process
- Appointment reminders
- Customer support

Pain Points

- Small fonts
- Complicated interfaces
- Confusing navigation
- Difficulty understanding online payments

- Password-related problems
- Dependence on family members
- Slow customer support
- Difficulty understanding technical or medical terminology

User Need

A senior citizen needs a simple, accessible, and easy-to-understand appointment application that allows them to independently book and manage appointments.

Problem Frame

Senior citizens need a simple and accessible hospital appointment system with clear navigation, readable information, and easy booking because complicated digital interfaces and online processes can make healthcare access difficult and may force them to depend on family members.

Recurring Expectations and Pain Points

After comparing the two user groups, the following common requirements and problems were identified.

Category	Busy Working Professional	Senior Citizen
Appointment Booking	Wants fast booking	Wants simple booking
Availability	Needs real-time slots	Needs clearly displayed slots
Navigation	Wants minimal steps	Needs simple navigation
Confirmation	Wants instant confirmation	Wants clear confirmation
Interface	Wants fast and responsive UI	Needs large and readable UI
Rescheduling	Needs easy rescheduling	Needs simple appointment management

Category	Busy Working Professional	Senior Citizen
Support	Wants quick customer support	Needs easily accessible assistance
Accessibility	Needs convenience	Needs accessibility features
Medical Information	Wants quick access	Needs easy access to records/prescriptions
Notifications	Wants reminders	Needs clear reminders

Major Recurring Pain Points

1. Complicated appointment booking processes
2. Lack of real-time appointment availability
3. Difficulty rescheduling appointments
4. Unclear appointment confirmation
5. Poor application performance
6. Difficulty navigating the application
7. Lack of accessibility features
8. Difficulty understanding online payment procedures
9. Dependence on others for assistance
10. Lack of clear reminders and notifications

Final Problem Frame

Patients need a hospital appointment system that is fast, reliable, simple, and accessible because different users face difficulties such as limited time, complicated navigation, lack of real-time availability, lengthy booking procedures, and difficulty using digital healthcare services independently.

Task 2: POV Statement Creation

- Consider one set of users and draft POV statements using the format: *User* + *Need* + *Insight*.
- Validate statements with peers to ensure clarity and relevance.

The POV statements are created using:

User + Need + Insight

POV Statement 1 – Busy Working Professional

User

A busy working professional who has limited free time because of work commitments.

Need

Needs a fast and reliable hospital appointment system that provides real-time doctor availability, instant confirmation, and easy rescheduling.

Insight

Unexpected work commitments can affect planned appointments, so the user needs to manage appointments quickly without wasting time.

Final POV Statement

A busy working professional needs a fast and reliable hospital appointment application with real-time doctor availability, instant confirmation, and easy rescheduling because limited free time and unexpected work commitments make lengthy appointment procedures difficult to manage.

POV Statement 2 – Senior Citizen

User

A senior citizen who may have difficulty using complex digital applications.

Need

Needs a simple and accessible hospital appointment application with large fonts, clear navigation, voice assistance, and easy booking.

Insight

Complicated interfaces, small text, and confusing online processes can make senior citizens dependent on family members for healthcare-related tasks.

Final POV Statement

A senior citizen needs a simple and accessible hospital appointment application with larger text, clear navigation, and assistance features because complicated interfaces and digital processes make it difficult to book appointments independently.

POV Statement Validation

The POV statements were reviewed to ensure that they:

- Clearly identify a specific user.
- Focus on a genuine user need.
- Explain the reason behind the need.
- Avoid directly prescribing a complete technical solution.
- Are based on user pain points and empathy insights.
- Can guide the next stage of Design Thinking, which is ideation.

Refinement After Validation

After reviewing the statements, the following improvements were made:

Initial framing:
"Patients need an easy hospital appointment app."

Refined framing:
"A busy working professional needs a fast and reliable hospital appointment application with real-time doctor availability, instant confirmation, and easy rescheduling because limited free time and unexpected work commitments make lengthy appointment procedures difficult to manage."

The refined statement is better because it identifies the specific user, specific needs, and underlying reason instead of making a general statement.

Task 3: Prepare Empathy Maps

- Prepare the empathy maps for each types of users as shown in the sample format.

Empathy Map 1 – Busy Working Professional

User: Priya Sharma

Age:		28
Occupation:	Software	Engineer
Location:		Mumbai

Technology Proficiency: High

The persona information is consistent with your Experiment 1, where Priya Sharma is identified as a busy software engineer who wants quick appointments, experienced doctors, short waiting times, reminders, and secure online payments.

Says

1. “I don't have much time to spend on booking an appointment.”
2. “I want to know which doctors are available right now.”
3. “I should be able to book an appointment in a few minutes.”
4. “I need confirmation immediately after booking.”
5. “I want an easy way to reschedule if something comes up at work.”

Thinks

1. “Will the appointment slot still be available when I finish booking?”
2. “I hope I don't have to go through too many steps.”
3. “What if I get an appointment but it conflicts with my work?”
4. “Can I trust the payment system?”
5. “I need to know exactly when my appointment is confirmed.”

Does

1. Searches for doctors using the application.
2. Compares doctor ratings and reviews.
3. Checks available appointment slots.
4. Books appointments online.
5. Reschedules appointments when work commitments change.

These behaviors are aligned with the behaviors identified for Priya in Experiment 1.

Feels

1. Stressed when the booking process is lengthy.

2. Frustrated when appointment slots are unavailable.
3. Anxious when confirmation is delayed.
4. Satisfied when an appointment is booked quickly.
5. Relieved when reminders and confirmation are received.

Key Insight

The busy working professional values speed, reliability, and convenience. Any unnecessary steps, delays, or uncertainty in the appointment process can cause frustration.

Empathy Map 2 – Senior Citizen

User: Rajesh Patel

Age:			65
Occupation:	Retired	Bank	Manager
Location:			Nashik
Technology Proficiency: Moderate			

Your Experiment 1 identifies Rajesh as a senior citizen who wants to book appointments without visiting the hospital and needs simple navigation, readable interfaces, voice assistance, and easy access to prescriptions.

Says

1. “The text should be large enough for me to read.”
2. “I don't understand complicated steps.”
3. “I want to book an appointment without asking someone for help.”
4. “I need clear instructions while making a payment.”
5. “I want to easily find my previous prescriptions.”

Thinks

1. “Am I selecting the correct doctor?”
2. “What if I make a mistake while booking?”
3. “Is my appointment really confirmed?”
4. “Will I be able to understand all the information?”
5. “What should I do if I forget my password?”

Does

1. Searches for nearby hospitals and doctors.
2. Reads doctor information carefully.
3. Checks available appointment slots.
4. Books appointments several days in advance.
5. Asks family members for help when the application becomes difficult to use.

These behaviors correspond to the persona created in Experiment 1.

Feels

1. Confused when navigation is complicated.
2. Frustrated when text is too small.
3. Anxious when making online payments.
4. Dependent when assistance from family members is required.
5. Confident when the application provides simple instructions.

Key Insight

The senior citizen values simplicity, clarity, accessibility, and confidence. A complicated interface can prevent independent use even when the required functionality is available.

Comparison of Empathy Maps

Aspect	Busy Working Professional	Senior Citizen
Main Concern	Saving time	Ease of use
Major Need	Fast booking	Simple booking
Main Pain Point	Lengthy processes	Complicated interface
Navigation	Wants fewer steps	Needs clear navigation
Appointment	Needs real-time availability	Needs clearly displayed availability

Aspect	Busy Working Professional	Senior Citizen
Confirmation	Wants instant confirmation	Wants clear confirmation
Assistance	Prefers self-service	May require assistance
Accessibility	Convenience-focused	Accessibility-focused
Emotional State	Frustrated by delays	Confused by complexity
Important Feature	Quick booking and rescheduling	Large fonts and simple navigation

Task 4: Refinement and Presentation

- Refine frames and POV statements based on feedback.
- Present them to the class or team, explaining how they connect to user needs.
- Document final versions for use in the next design thinking stage (Ideation).

Refined Problem Statement 1

Busy working professionals need a fast and reliable hospital appointment system that allows them to search for doctors, view real-time availability, book appointments, receive instant confirmation, and reschedule appointments because their work schedules provide limited free time and unexpected commitments can interfere with planned appointments.

Refined Problem Statement 2

Senior citizens need a simple and accessible hospital appointment application that provides large readable text, clear navigation, easy booking, and assistance features because complicated interfaces and digital processes can make healthcare access difficult and reduce their ability to use the system independently.

Final POV Statements

POV 1 – Busy Working Professional

A busy working professional needs a fast and reliable hospital appointment application with real-time doctor availability, instant confirmation, and easy rescheduling because

limited free time and unexpected work commitments make lengthy appointment procedures difficult to manage.

POV 2 – Senior Citizen

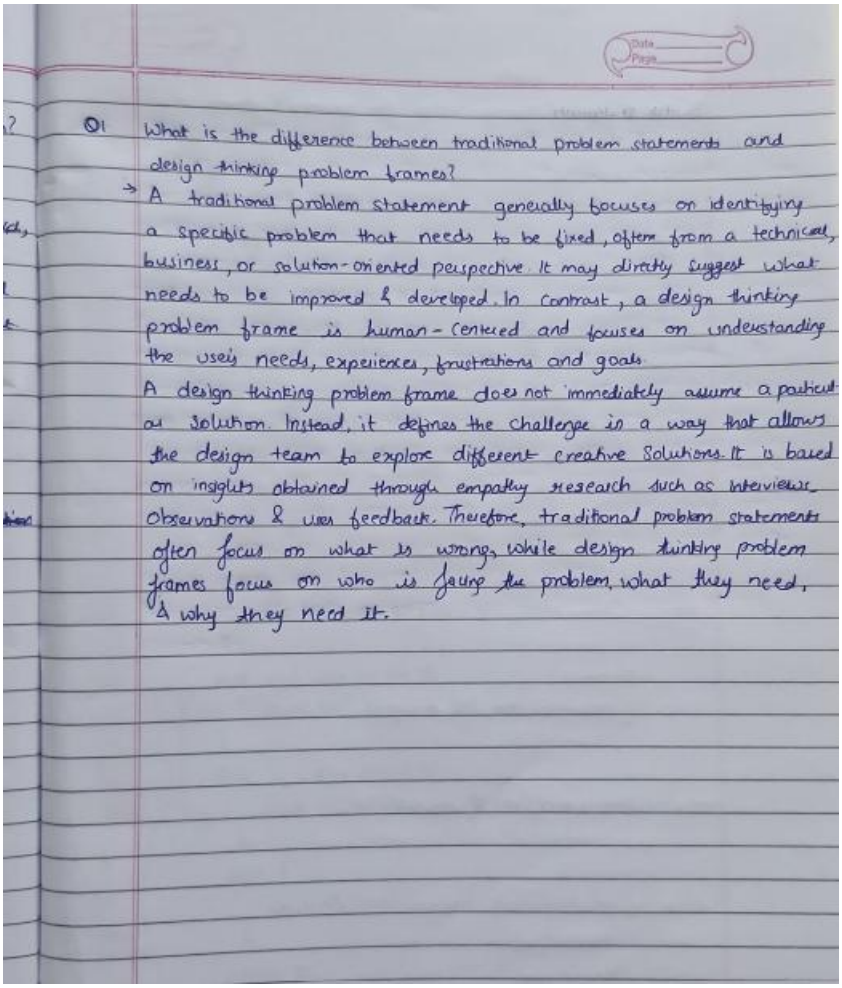
A senior citizen needs a simple and accessible hospital appointment application with larger text, clear navigation, and assistance features because complicated interfaces and digital processes make it difficult to book appointments independently.

These final POV statements can be used as the foundation for the Ideation stage, where possible solutions and features will be generated.

Post Laboratory Questions:

Following 4 questions are to be handwritten and attach photograph of the answer in this document.

1. What is the difference between traditional problem statements and design thinking problem frames?

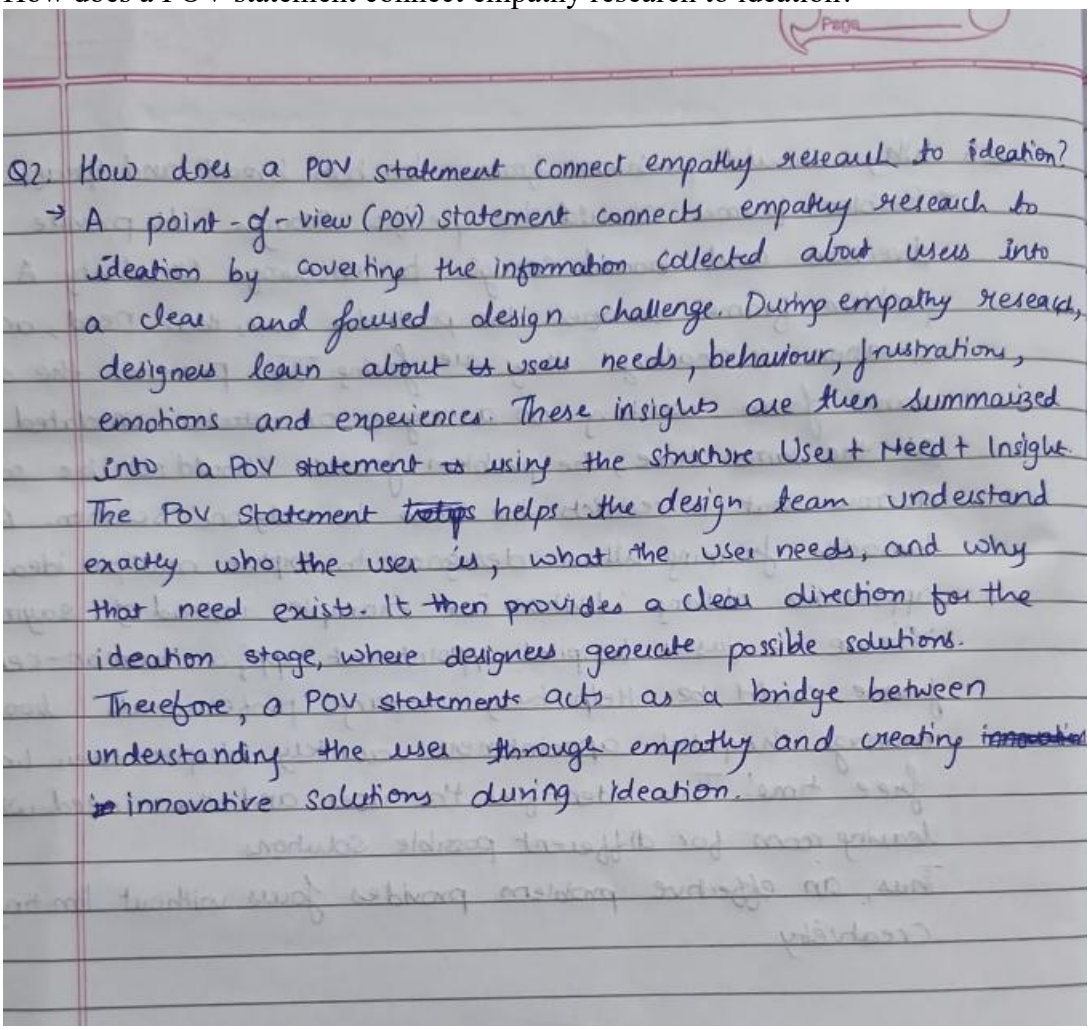


Q1 What is the difference between traditional problem statements and design thinking problem frames?

→ A traditional problem statement generally focuses on identifying a specific problem that needs to be fixed, often from a technical, business, or solution-oriented perspective. It may directly suggest what needs to be improved & developed. In contrast, a design thinking problem frame is human-centered and focuses on understanding the user's needs, experiences, frustrations and goals.

A design thinking problem frame does not immediately assume a particular solution. Instead, it defines the challenge in a way that allows the design team to explore different creative solutions. It is based on insights obtained through empathy research such as interviews, observations & user feedback. Therefore, traditional problem statements often focus on what is wrong, while design thinking problem frames focus on who is facing the problem, what they need, & why they need it.

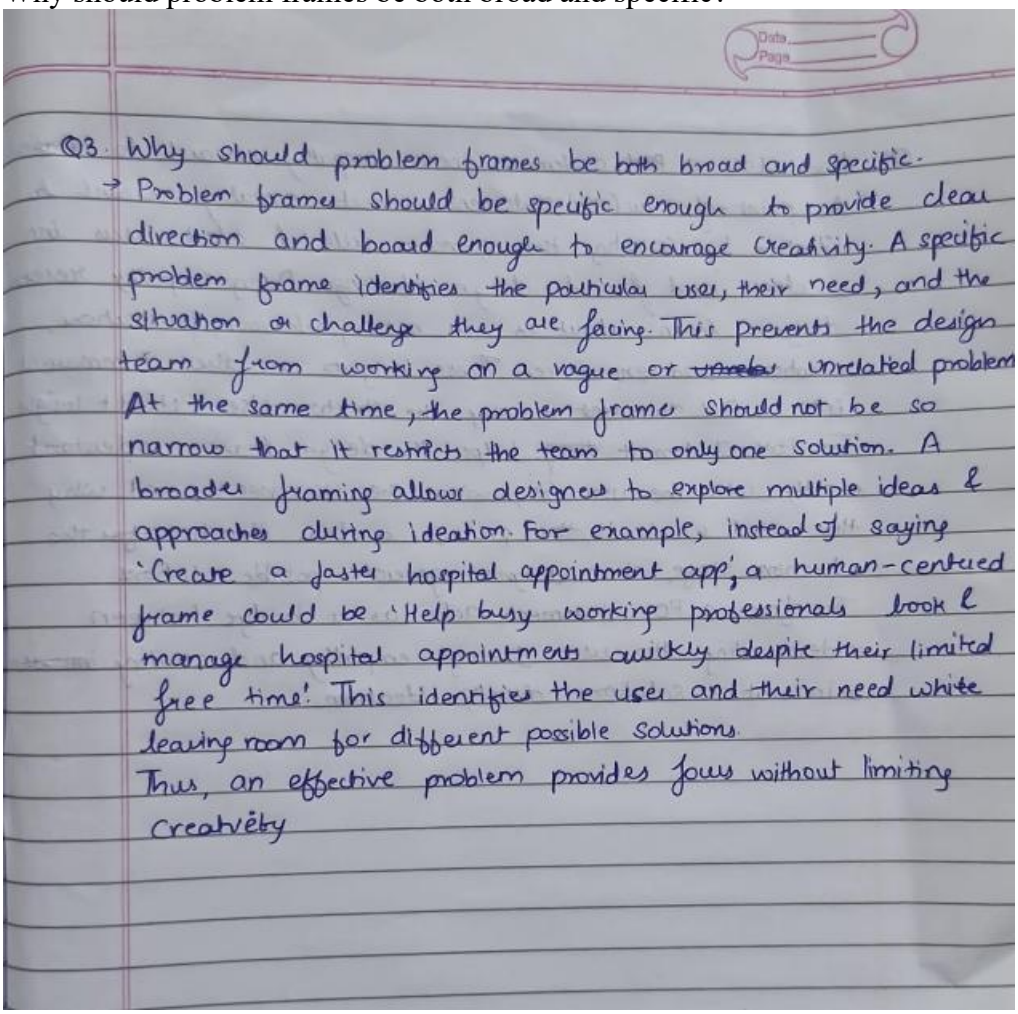
2. How does a POV statement connect empathy research to ideation?



Q2. How does a POV statement connect empathy research to ideation?

→ A point-of-view (POV) statement connects empathy research to ideation by converting the information collected about users into a clear and focused design challenge. During empathy research, designers learn about ~~it~~ users needs, behaviour, frustrations, emotions and experiences. These insights are then summarized into a POV statement ~~is~~ using the structure User + Need + Insight. The POV Statement ~~helps~~ helps the design team understand exactly who the user is, what the user needs, and why that need exists. It then provides a clear direction for the ideation stage, where designers generate possible solutions. Therefore, a POV statements acts as a bridge between understanding the user through empathy and creating ~~innovative~~ innovative solutions during ideation.

3. Why should problem frames be both broad and specific?

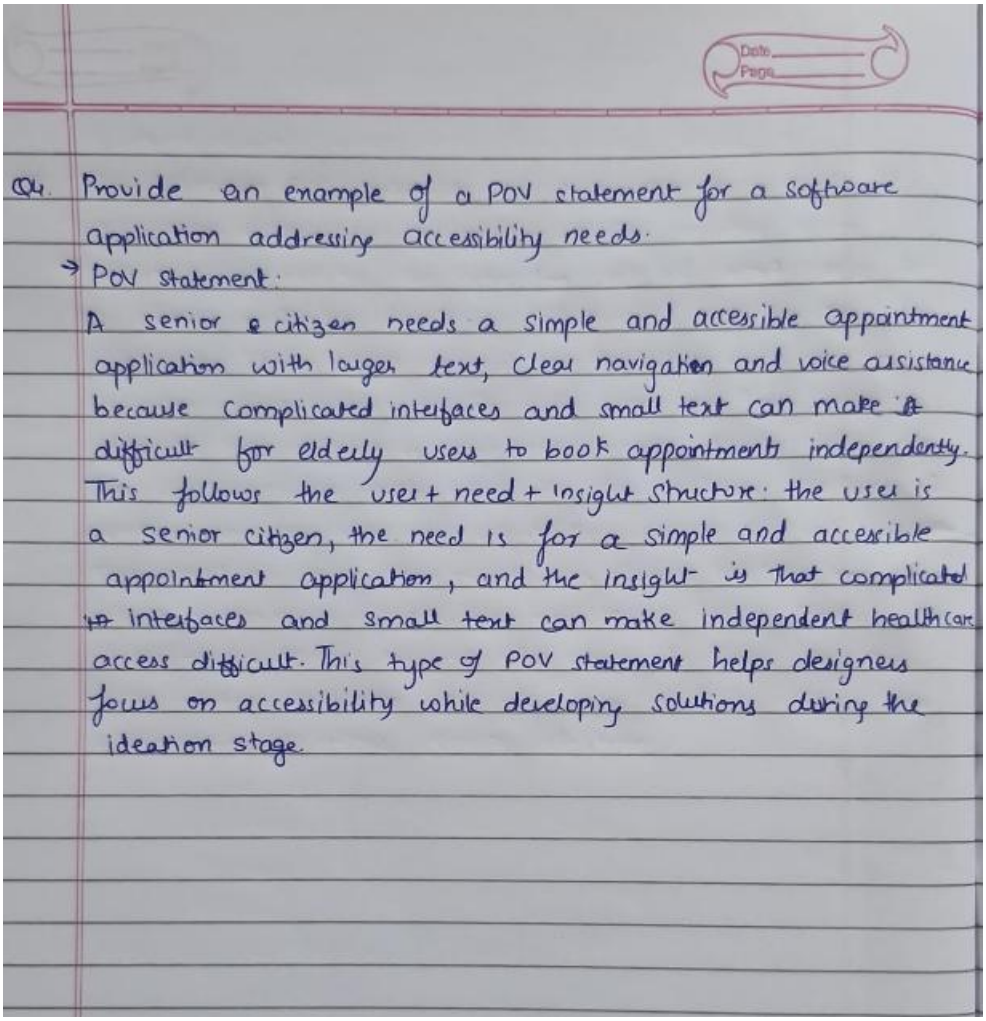


Q3. Why should problem frames be both broad and specific.

→ Problem frames should be specific enough to provide clear direction and board enough to encourage creativity. A specific problem frame identifies the particular user, their need, and the situation or challenge they are facing. This prevents the design team from working on a vague or ~~vague~~ unrelated problem. At the same time, the problem frame should not be so narrow that it restricts the team to only one solution. A broader framing allows designers to explore multiple ideas & approaches during ideation. For example, instead of saying 'Create a faster hospital appointment app', a human-centered frame could be 'Help busy working professionals book & manage hospital appointments quickly despite their limited free time'. This identifies the user and their need while leaving room for different possible solutions.

Thus, an effective problem provides focus without limiting creativity.

4. Provide an example of a POV statement for a software application addressing accessibility needs.



Q4. Provide an example of a POV statement for a software application addressing accessibility needs.

→ POV statement:

A senior citizen needs a simple and accessible appointment application with larger text, clear navigation and voice assistance because complicated interfaces and small text can make it difficult for elderly users to book appointments independently. This follows the user + need + insight structure: the user is a senior citizen, the need is for a simple and accessible appointment application, and the insight is that complicated interfaces and small text can make independent healthcare access difficult. This type of POV statement helps designers focus on accessibility while developing solutions during the ideation stage.